ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Technical Programming paper 2 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Award marks for correct logic even if there are small syntax slips. Accept any correct equivalent code and variable names.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 C ✓ 1.3.3 B ✓ 1.3.4 H ✓ 1.3.5 D ✓ 1.3.6 F ✓ 1.3.7 E ✓ 1.3.8 G ✓ 1.3.9 I ✓ 1.3.10 J ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 #include <iostream> and namespace ✓, variables as double/float ✓, prompt with cout ✓, cin ✓, correct formula ✓, output with label ✓ (6)

```
#include <iostream>
using namespace std;

int main() {
    double c, f;
    cout << "Temperature in Celsius: ";
    cin >> c;
    f = c * 9 / 5 + 32;
    cout << "Temperature in Fahrenheit: " << f << endl;
    return 0;
}
```

2.2 switch on the character ✓, four cases with correct messages ✓✓✓✓ (1 each), break statements ✓, default case ✓ (7)

```
switch (command) {
    case 'F': cout << "Move forward"; break;
    case 'B': cout << "Move back";    break;
    case 'L': cout << "Turn left";    break;
    case 'R': cout << "Turn right";   break;
    default:  cout << "Unknown command";
}
```

2.3 It is easier to read and maintain when one variable is compared with many fixed values ✓; it avoids long chains of else-if and can be faster / clearer ✓. (2)

[15]

QUESTION 3

3.1.1 Start values ✓, n = 1 sum 1 ✓, n = 3 sum 10 ✓, n = 5 sum 35 ✓, n = 7 sum 84 and loop ends at n = 9 ✓ (5)

n (start of pass)	sum after the pass	n after the pass
1	1	3
3	10	5
5	35	7
7	84	9 (loop stops)

3.1.2 84 ✓ (1)

3.2 Counter and total set to 0 ✓, first reading before the loop (or loop structure that checks −1) ✓, while (reading != −1) ✓, add to total and count ✓✓, read next value inside loop ✓, avoid division by zero ✓, display count and average ✓ (8)

```
double reading, total = 0;
int count = 0;
cout << "Enter a reading (-1 to stop): ";
cin >> reading;
while (reading != -1) {
    total = total + reading;
    count++;
    cout << "Enter a reading (-1 to stop): ";
    cin >> reading;
}
if (count > 0) {
    cout << count << " readings, average " << total / count << endl;
} else {
    cout << "No readings entered" << endl;
}
```

[14]

QUESTION 4

4.1 Return type ✓, name and two parameters ✓, return length × width ✓, correct braces ✓, call with arguments ✓, result displayed / stored ✓ (6)

```
double area(double length, double width) {
    return length * width;
}

int main() {
    double a = area(30, 20);
    cout << "Area: " << a << " cm2" << endl;
    return 0;
}
```

4.2 A local variable is declared inside a function and exists only there ✓; a global variable is declared outside all functions and can be used anywhere ✓; any function can change a global variable, which makes errors hard to find ✓. (3)

4.3.1 33 ✓ (1)

4.3.2 Start with the first element as the smallest ✓, loop through the array ✓✓, compare ✓, update smallest ✓, display ✓ (6)

```
int smallest = dist[0];
for (int i = 1; i < 6; i++) {
    if (dist[i] < smallest) {
        smallest = dist[i];
    }
}
cout << "Smallest distance: " << smallest << " cm";
```

4.3.3 Counter set to 0 ✓, loop through all elements ✓, condition < 20 ✓, increase counter and display ✓ (answer 2) (4)

```
int close = 0;
for (int i = 0; i < 6; i++) {
    if (dist[i] < 20) close++;
}
cout << close;   // displays 2
```

[20]

QUESTION 5

5.1.1 Attributes: name, battery ✓; methods: Robot (constructor) ✓, drive, getBattery ✓ (3)

5.1.2 Create object with the name ✓, call drive(15) ✓, call getBattery() ✓, display with cout ✓ (4)

```
Robot rover("Rover1");
rover.drive(15);
cout << rover.getBattery() << "%" << endl;
```

5.1.3 100 − 15 × 2 = 70 (%) ✓ (1)

5.1.4 Encapsulation – the data can only be changed through the class's methods, so it cannot be set to wrong values from outside ✓. (1)

5.2 forever block ✓; set digital pin 13 output HIGH, wait 0,5 s ✓; set digital pin 13 output LOW, wait 0,5 s ✓ (accept 1 s on / 1 s off if clearly once per cycle) (3)

[12]

QUESTION 6

6.1 Error 1: missing semicolon after using namespace std ✓ – add ; ✓. Error 2: i < 10 stops at 9 ✓ – change to i <= 10 ✓. Error 3: capital I is undeclared (C++ is case sensitive) ✓ – change to i ✓ (6)

6.2 Any THREE ✓✓✓: breakpoints; stepping line by line (step into / over); watching variable values (watch window); viewing the call stack; inspecting output / messages (3)

[9]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE